Vignesh S Suvarna

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Portfolio | GitHub | LinkedIn | Codeforces

Summary

Computer Science Engineering student at Atria Institute of Technology with hands-on experience in Python, C++, JavaScript, SQL, React.js, Node.js, and REST APIs. Built full-stack, AI-powered, and Android applications, with coursework in object-oriented programming, database systems, and operating systems.

Projects

NAYIDISHA | *React, Node.js, Express, PostgreSQL, ESP32, Vercel* **2026 – Present**

- Built a full-stack platform for SAMA Foundation to support care management, progress tracking, and personalized education for families and caretakers.
- Designed secure REST APIs with Node.js, Express, and PostgreSQL, enforcing role-based access control for parents, caretakers, and administrators.
- Integrated an ESP32-based IoT module that collects environmental data to adapt learning content; deployed the web app on Vercel.

HERMES | *Python, PyQt5, Groq API, ChromaDB, PyAutoGUI, SQLite* **2026 – Present**

- Developed a Python desktop assistant with a PyQt5 interface for workflow and GUI automation.
- Integrated the Groq API for low-latency LLM responses, and used ChromaDB and SQLite for semantic memory and persistent context.
- Implemented screen analysis, speech recognition, web automation, and real-time monitoring features.

Hearing Aid Pro | *Kotlin, Android SDK, Drizzle ORM, XML, SharedPreferences, Custom Canvas* **2026 – Present**

- Developed a low-latency Android hearing assistant in Kotlin, using native audio APIs for real-time audio processing.
- Implemented DSP-based noise cancellation and adaptive audio filtering to improve speech clarity in noisy environments.
- Added real-time audio visualization using a custom Canvas and personalized hearing profiles stored locally.

HireSphere | *React, Node.js, Express, PostgreSQL, Drizzle ORM, Tailwind CSS, Zustand* **2026 – Present**

- Architected a full-stack campus placement platform for managing student portfolios and placement analytics.
- Developed secure REST APIs with Node.js, Express, Neon PostgreSQL, and Drizzle ORM, using JWT authentication and RBAC.
- Built React and Tailwind CSS dashboards with Zustand state management, ATS resume screening, and placement analytics.

Education

Atria Institute of Technology **Bengaluru, Karnataka**
Bachelor of Engineering in Computer Science and Engineering | CGPA: 8.29/10 *Expected Graduation: 2028*

Pre-University Course (II PUC) **2024**
Percentage: 91%

Secondary School Leaving Certificate (SSLC) **2022**
Percentage: 92.48%

Technical Skills

Languages: C, C++, Python, JavaScript, HTML, CSS, SQL
Technologies & Frameworks: React.js, Node.js, Express.js, Vite, Tailwind CSS, PostgreSQL, MySQL, SQLite, REST APIs, LLMs, RAG
Developer Tools: VS Code, Git, GitHub, Vercel, Postman, Android Studio
Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, REST API Design

Achievements

- Presented ArgiConnect, an original project concept, at the New Horizon Ideathon 2025.
- Developed Smart Student Hub, an application for student resources and engagement, for the Smart India Hackathon internal round.
- Leading the development of an Android-based hearing assistance solution in collaboration with the Chandrashekhar Institute of Speech and Hearing.

Relevant Coursework

Completed: Object-Oriented Programming, Database Management Systems, Operating Systems
Currently Studying: Data Structures and Algorithms, Computer Networks, Machine Learning